

Endogenous Trade Fragmentation*

[Author]

July 2026

Abstract

Why might trade fragmentation feed on itself? I model firms that can keep a cheap cross-bloc supplier or pay a premium for a politically safe one, and I derive the feedback that makes their choices strategic complements: trade is its own hostage. Escalation of a geopolitical dispute into supply disruption destroys the surplus on remaining cross-bloc trade, so each firm that friend-shores erodes the deterrent protecting every firm that stays. Two further channels, thinning trade infrastructure and scale economies in friendly capacity, compound the first. The model delivers a fragmentation multiplier hidden from cross-sectional evidence, and a fold bifurcation in which critical sectors tip while aggregate trade barely moves. It also delivers hysteresis: a twelve-year confrontation leaves the network permanently fragmented at a dispute intensity that, had it never occurred, would have left the network intact. In a calibration anchored to the post-2022 environment, the critical sector sits about two thirds of the way to its tipping point, between half and all of the way across the sensitivity grid, and the corrective wedge below the fold is a subsidy on *maintaining* cross-bloc trade of six percent ad valorem: at this calibration, friend-shoring subsidies have the wrong sign. A subsidy paying a quarter of the switching premium tips the sector and more than doubles the disruption hazard it was meant to insure against. Stockpiles buy resilience while preserving the hostage and push the tipping point away. Past the fold the wedge flips sign: defend the bridge, then burn it.

JEL codes: F13, F51, F52, D85, C62.

Keywords: fragmentation, friend-shoring, supply chains, strategic complementarity, tipping points, deterrence.

1 Introduction

China supplied about 22 percent of United States goods imports in 2017 and about 14 percent in 2023. Total imports stayed high over those years. The gap moved to Vietnam, Mexico, and other countries positioned between the blocs (Alfaro and Chor, 2023; Freund et al., 2023). The movement was uneven: trade between geopolitical blocs has slowed relative to trade within them

*Revised draft: July 2026. The empirical sections specify the product-level trade, input-output, and sanctions data used for estimation. Every number in the text is either documented in a cited source or an output of the model solved here. Replication code in Julia and R accompanies the draft.

since Russia invaded Ukraine in February 2022, with the sharpest breaks in strategically sensitive products (Gopinath et al., 2025). The reallocation is a rewiring of the network along political lines, concentrated exactly where substitution is hardest.

The question of this paper is whether that rewiring is self-limiting or self-reinforcing: the smooth repricing of a one-time increase in geopolitical risk, or the early stage of an unraveling in which each firm’s exit makes exit more attractive for the next. The distinction matters for more than forecasting. Governments are committing tens of billions of dollars to friend-shoring subsidies on the theory that private firms undervalue resilience, and the quantitative literature prices fragmentation counterfactuals conditional on an imposed split, with costs ranging from 0.2 to 7 percent of global GDP by depth (Aiyar et al., 2023; Baqaee et al., 2024; Attinasi et al., 2023). Whether the split is something that happens *to* the trade network or something the network does to itself changes both the forecast and the sign of good policy. Existing work treats the risk as exogenous. The contribution here is to make it an equilibrium object.

The model is deliberately spare. In each sector, firms currently sourcing a key input across the bloc boundary choose between keeping the cheap supplier, and bearing a disruption hazard, or switching to a politically safe supplier at an idiosyncratic cost premium. What makes the choices interact is the hazard, and here I derive the feedback rather than assume it. Escalating a geopolitical dispute into export controls or embargo destroys the surplus on whatever cross-bloc trade remains, and a government weighs that cost against the political benefit of escalating, so the probability that any given link is severed falls with the volume of trade still crossing the boundary. Trade is its own hostage. The evidence that commerce deters conflict is old and quantitatively serious (Martin et al., 2008; Jackson and Nei, 2015). I hand that deterrence back to the firms as an externality. Each firm that friend-shores shrinks the hostage, and the smaller hostage raises the hazard on every remaining link, pushing its neighbors toward the same exit. Two further channels compound the feedback and I model them explicitly: disruptions last longer when fewer firms share routes and insurance, and the premium for safe suppliers falls as friendly capacity scales. The first channel is a security externality, the other two are familiar thick-market forces, and the model signs and separates all three.

Three results organize the analysis. The first is an amplification result with an econometric sting. Because choices are complements, any shock to the perceived hazard moves the equilibrium by a multiple $1/(1 - \Phi')$ of its direct effect, where Φ' is the slope of the aggregate best response: a fragmentation multiplier of 1.25 at the calibrated post-2022 point, and unboundedly large near a tipping point. The sting is that cross-sectional comparisons, across firms, links, or sectors within the same equilibrium, identify only the direct effect: the feedback is a common term that differences out, so the reduced-form literature on supplier switching measures the numerator of a ratio whose denominator is the object of interest. Only aggregate moments, or the structural moments I specify in Section 6, can recover the multiplier.

The second result is that tipping, when it comes, is sectoral and hysteretic. The disruption loss a firm prices is the cost jump from losing an input, which the CES structure makes large exactly where substitution elasticities are low and cost shares high, so low-elasticity, high-centrality

sectors reach their fold bifurcation at dispute intensities that leave flexible sectors indifferent. At the calibration, the critical sector’s fold sits at a dispute hazard of 13.1 percent per year against a post-2022 level of 10 percent: the 2022 reassessment consumed about 66 percent of the distance, a share that runs from 47.0 to 100.0 percent across the deterrence and capacity parameters spanned in the sensitivity analysis. Past the fold the sector jumps from single-digit to near-complete fragmentation while aggregate trade barely registers the event, which is why aggregate statistics are the wrong early-warning instrument. And the fold carries a band of hysteresis: in the model’s central experiment, a twelve-year confrontation raises the dispute hazard temporarily and then subsides, and the sector that lived through it ends 90 percent fragmented while an identical sector that entered directly at the terminal hazard ends at 12 percent. Each firm could return at will. The trap is an equilibrium rather than a friction, and inside it the disruption hazard itself has doubled: fragmentation raises the probability of the very event it insures against, which is the security dilemma in equilibrium form.

The third result is about policy, and it reverses the current default. The deterrence channel makes private friend-shoring socially excessive: the firm that exits counts its own premium and omits the hazard it raises for everyone remaining. The corrective instrument below the fold is therefore a subsidy on *maintaining* cross-bloc trade, worth 6.4 percent ad valorem at the calibrated post-2022 point, and the friend-shoring subsidies actually in force have the opposite sign. The model prices the mistake: a subsidy paying one quarter of the switching premium moves the critical sector’s fold from 13.1 to 9.7 percent, below the current dispute hazard, tipping the sector to 94 percent fragmentation and raising the equilibrium disruption hazard from 4.5 to 9.9 percent per year. Resilience policy of this form purchases the crisis it was bought to insure against. A stockpile of equal ambition, which cuts the loss a disruption inflicts while holding the hostage fixed, does the opposite: it pushes the fold out to 19.7 percent and lowers the hazard. And past the fold the prescription flips sign, from defending the remaining trade to accelerating the transition, because once the hostage is spent the scale externality among switchers is all that is left: defend the bridge, then burn it.

The paper draws on three literatures and differs from each in one identified way. The resilience-policy literature, best represented by Grossman et al. (2023), studies diversification against exogenous disruption risk in unique-equilibrium environments. My disruption risk is endogenous to the aggregate sourcing pattern, which is where the multiplicity and the sign reversal come from. The network-fragility literature, notably Elliott et al. (2022), generates discontinuous fragility from endogenous link formation, but its propagation risks are statistical rather than strategic. Here the disruption probability is chosen by a government whose restraint depends on what the network still carries, which is also what makes the externality Pigouvian rather than purely informational. And the trade-and-conflict literature from Martin et al. (2008) to Jackson and Nei (2015) estimates the deterrence relation I build on but leaves firm anticipation out: closing that loop, from deterrence to portfolio choice back to deterrence, is precisely what generates tipping. The measurement papers that motivated this one, Gopinath et al. (2025), Fernández-Villaverde et al. (2024), and Alfaro and Chor (2023), document the

facts the model is built to explain: reallocation along political lines, concentrated in strategic sectors and gradual in aggregate. The geoeconomics framework of Clayton et al. (2023) and the weaponized-interdependence thesis of Farrell and Newman (2019) study the hegemon’s use of network power. I study the network’s anticipatory response, which is the other half of that equilibrium. Standard decoupling counterfactuals are nested exactly: shutting all three feedback channels recovers the exogenous-risk model in which fragmentation is linear in shocks and policy is a level choice.

The quantitative exercise is disciplined in three ways. The welfare accounting maps input-cost burdens to GDP through cross-bloc trade shares, leaving variety and diffusion gains to the measured extensions, which puts the aggregate costs at the bottom of the published range. Section 4 prices an additional government value v of independence and shows how large it must be before the sign of the corrective wedge changes. Section 6 maps every parameter to a public-data moment, so the calculations below and the empirical design use the same objects.

My plan is as follows. Section 2 sets out the environment and derives the feedback. Section 3 characterizes equilibria, multipliers, tipping, and hysteresis. Section 4 treats welfare and policy. Section 5 calibrates and reports the quantitative experiments. Section 6 specifies the empirical designs. Section 7 concludes. Proofs and extensions are in the appendix.

2 The Model

The model separates analytical structure from quantitative implementation. The threshold characterization (Lemma 1), the decomposition of the complementarity (Proposition 1), the multiplier formula and the micro-macro gap (Proposition 2, Corollary 1), and the ordering and hysteresis results (Propositions 3 and 4) are analytical. The multiplier exceeds one where the feedback channels operate. Folds and traps occur where the sector map’s slope crosses one, and Section 5 locates that condition in the post-2022 calibration. The uniform benefit distribution gives the exact linear hazard in (2). Log-concave benefit distributions deliver the same monotonicity. Every quantitative number below is tied to a named input and then mapped to an empirical moment in Section 6.

2.1 Firms and the sourcing choice

There are two blocs and M sectors. In sector m , a unit mass of Home firms currently source a key intermediate input from a cross-bloc supplier, and sector m carries a share w_m of cross-bloc intermediate trade. Time is discrete and annual. Each firm chooses a sourcing mode: *cross-bloc*, keeping the incumbent supplier, or *friend-shored*, replacing it with an in-bloc supplier of the same input.

Friend-shoring costs firm i a flow premium g_i , expressed ad valorem on the input: the in-bloc supplier is dearer or must first be built. The premium is idiosyncratic, $g_i \sim H_m$, lognormal with median $\bar{g}_m$ and log dispersion s . Heterogeneity stands in for relationship capital and product-specific reshoring costs. A fraction f_m^0 of firms switched before geopolitics entered the

calculus, for wages or diversification, and I take that margin as given.

Cross-bloc sourcing is cheap but exposed. With hazard π_m per year the link is severed by policy: an export control or an embargo. A severed firm loses λ_m , the disruption loss expressed as a flow-equivalent ad valorem cost, and then friend-shores perforce. The loss has a standard CES anatomy. If the firm’s input bundle aggregates the cross-bloc variety and substitutes with elasticity σ_m , and the variety carries cost share s_m , losing it raises the input cost index by the factor $(1 - s_m)^{1/(1-\sigma_m)}$, so a disruption lasting D_m years costs

$$\lambda_m^0 = \left[(1 - s_m)^{\frac{1}{1-\sigma_m}} - 1 \right] D_m, \quad (1)$$

large precisely where substitution is hard (σ_m low), exposure deep (s_m high), and recovery slow (D_m long). Equation (1) is what makes “criticality” a parameter rather than an adjective, and it is measurable sector by sector from substitution elasticities and input-output tables (Broda and Weinstein, 2006; OECD, 2026). The disruption studies of Barrot and Sauvagnat (2016), Boehm et al. (2019), and Carvalho et al. (2021) document both the losses and recovery times on the order of a year for specific inputs.

The sourcing choice reduces exactly to a threshold rule.

Lemma 1 (The flow comparison is exact). *Suppose a disrupted firm switches perforce and firms discount at any $\beta_f \in (0, 1)$. In a stationary environment, staying is optimal if and only if the flow comparison holds:*

$$\pi_m \lambda_m \leq g_i^{\text{eff}},$$

where g_i^{eff} is the effective premium defined below. Forward-looking and myopic thresholds coincide.

The logic is that switching is an absorbing state whose cost is the premium forever, while staying pays the expected disruption loss and, upon disruption, inherits the same absorbing state, so the continuation values cancel and the flow comparison survives intact (Appendix A). Along transition paths a small option-value correction appears. Appendix C bounds it and connects the selection issue to Krugman (1991) and Matsuyama (1991).

2.2 The feedback, derived

Let $F_m \in [0, 1]$ denote the fragmented share of sector m ’s cross-bloc trade, and $F = \sum_m w_m F_m$ the aggregate. Three objects in the firm’s comparison depend on them.

The hazard: trade is its own hostage. Disputes between the blocs arrive with probability δ_m per year. When one arrives, the adversary government chooses whether to escalate it into supply disruption against sector m . Escalation yields a political benefit b , drawn uniformly on $[0, \bar{b}]$, and destroys the surplus on the trade it severs: a cost $\vartheta_m \bar{S}_m (1 - F_m)$ from the sector’s remaining cross-bloc trade, plus $\vartheta_a \bar{S} (1 - F)$ from the broad retaliation and reputational damage that scale with total interdependence. The government escalates when the benefit clears the cost,

so the link-level hazard is

$$\pi_m(F_m, F) = \delta_m [1 - \theta(1 - F_m)] + \eta_a F, \quad (2)$$

with $\theta = \vartheta_m \bar{S}_m / \bar{b}$ the share of escalations deterred by an intact sector and $\eta_a = \delta_m \vartheta_a \bar{S} / \bar{b}$ the systemic term (Lemma 2, Appendix B). The linearity comes from the uniform benefit distribution, and any log-concave distribution delivers the same monotonicity. Equation (2) is the paper's engine: the hazard each firm prices falls with the trade that remains, so every exit raises the hazard for everyone else. The deterrence relation itself is the one Martin et al. (2008) estimate across a century of dyadic data. What is new is handing it back to the firms as an externality.

The loss: routes thin as trade leaves. Shared infrastructure spreads disruption: alternative routings, insurance pools, spot markets, and buffer inventories are thick when many firms use the corridor and thin when few do. I let the loss rise linearly with the sector's fragmentation,

$$\lambda_m(F_m) = \lambda_m^0 (1 + \xi F_m), \quad (3)$$

with $\xi \geq 0$ the thinning elasticity.

The premium: friendly capacity scales. The in-bloc supply base gets cheaper as it gets bigger: entry, learning, and fixed-cost spreading in the destination countries compress the premium. The effective premium is

$$g_i^{\text{eff}} = g_i (1 - \chi F_m), \quad (4)$$

with $\chi \geq 0$ the capacity-scale elasticity. This is the channel the post-2018 reallocation to Vietnam and Mexico makes visible (Alfaro and Chor, 2023): each cohort of switchers builds the capacity that lowers the next cohort's cost.

Combining (2)–(4) with Lemma 1, firm i friend-shores when g_i falls below the threshold

$$\hat{g}_m(F_m, F) = \frac{\pi_m(F_m, F) \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m}, \quad (5)$$

and the sector's fragmented share consistent with individual choices is

$$F_m = T_m(F_m, F) \equiv f_m^0 + (1 - f_m^0) H_m(\hat{g}_m(F_m, F)). \quad (6)$$

Proposition 1 (Strategic complementarity, decomposed). *The threshold $\hat{g}_m$ is strictly increasing in F_m and F , with*

$$\frac{\partial \hat{g}_m}{\partial F_m} = \underbrace{\frac{\delta_m \theta \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m}}_{\text{deterrence erosion}} + \underbrace{\frac{\pi_m \lambda_m^0 \xi}{1 - \chi F_m}}_{\text{route thinning}} + \underbrace{\frac{\pi_m \lambda_m^0 (1 + \xi F_m) \chi}{(1 - \chi F_m)^2}}_{\text{capacity scale}} > 0,$$

so friend-shoring decisions are strategic complements within and across sectors. The complementarity vanishes if and only if $\theta = \xi = \chi = \eta_a = 0$, in which case the model reduces to the standard exogenous-risk sourcing problem with a unique equilibrium that is linear in shocks.

The decomposition matters because the three channels have different owners and different policy handles. Deterrence erosion is a security externality: it runs through a government's escalation calculus and exists even if markets for routes and capacity are perfectly competitive. Thinning and scale are pecuniary-cum-technological externalities of the kind familiar since Cooper and John (1988). Section 6 assigns each channel its own empirical moment, because their relative size decides both whether tipping is reachable and what the right instrument is.

The static objects are now in place: a threshold economy in each sector, coupled through the hazard and the aggregate. What remains is to characterize the fixed points of (6), and that is where the multiplier and the tipping structure live.

3 Equilibrium: Multipliers, Tipping, Hysteresis

3.1 Structure and the fragmentation multiplier

An equilibrium is a profile $(F_1, \dots, F_M)$ solving (6) for every sector, with $F = \sum_m w_m F_m$. The cross-sector coupling η_a is quantitatively minor at the calibration, so the exposition here treats sectors one at a time and the computations keep the full system.

Proposition 2 (Equilibrium set and the multiplier). *T_m maps $[f_m^0, 1]$ into itself, continuously and increasingly, so equilibria exist. Generically they are finite in number, odd, and alternate stable and unstable, stability meaning $\Phi'_m \equiv dT_m/dF_m < 1$ at the fixed point. At any stable equilibrium, the response of F_m^* to a small shift dz in any exogenous component of the threshold is*

$$\frac{dF_m^*}{dz} = \frac{(1 - f_m^0) h_m(\hat{g}_m) \partial \hat{g}_m / \partial z}{1 - \Phi'_m} = M_m \times (\text{direct effect}), \quad M_m = \frac{1}{1 - \Phi'_m} > 1,$$

and $M_m \rightarrow \infty$ as the equilibrium approaches a fold.

Corollary 1 (Micro and macro). *Comparisons across firms, links, or sectors within an equilibrium identify the direct effect: the feedback term Φ'_m is common to all units and differences out of any cross-sectional or difference-in-differences estimate. The equilibrium response exceeds the measured response by the factor M_m , which aggregate variation recovers, and which is largest exactly when the question matters most.*

The corollary should unsettle the growing reduced-form literature on supplier switching, including the designs I propose in Section 6. A well-identified estimate of how much more exposed firms switched after 2022 is the numerator of a ratio. The denominator, $1 - \Phi'_m$, is invisible in the cross section because every firm in the sector faces the same F_m . At the calibrated post-2022 point the gap is modest, $M_c = 1.25$ in the critical sector. The point is that its size is a diagnosis of distance from tipping, and only aggregate or structural moments can deliver it.

3.2 Folds and the sectoral cascade

Proposition 3 (Folds under common non-loss primitives). *(i) The stable low branch $F_m^{lo}(\delta)$ is increasing in the dispute hazard δ . If $\max_F \Phi'_m > 1$ at some δ , there is a band $[\delta_{lo}^*, \delta_{hi}^*]$ on which three equilibria coexist, and at δ_{hi}^* the low branch is destroyed in a fold and the sector jumps discontinuously to the high branch. (ii) Across sectors identical except for the loss λ_m^0 , both fold points are strictly decreasing in λ_m^0 : sectors tip in the order of (1), lowest substitution elasticity, deepest exposure, slowest recovery first. (iii) The aggregate inherits a jump of size $w_m (F_m^{hi} - F_m^{lo})$ at each sectoral fold: when tipping sectors are small, aggregate fragmentation moves smoothly through events that are discontinuous sector by sector.*

The empirical implementation computes fold hazards sector by sector using the full vector $(\lambda_m^0, H_m, \theta_m, \xi_m, \chi_m, f_m^0)$. Proposition 3 isolates the λ_m^0 component: holding the other non-loss primitives fixed, criticality orders the folds. Premium dispersion then determines how sharp the fold is. The condition $\max_F \Phi'_m > 1$ requires the premium density to be high where the threshold moves, so enough heterogeneity smooths the fold. At the calibrated log dispersion of 0.5 the critical sector's fold band is [11.4, 13.1] percent. Recomputing at dispersion 0.4 and 0.3 widens the band, and raising dispersion to 0.7 log points removes the fold. Premium dispersion is therefore load-bearing, and Design 4 estimates it directly from unit-value data.

Part (iii) is the measurement implication. A critical sector carrying fifteen percent of cross-bloc trade can pass its fold while the aggregate series registers a few percentage points spread over a decade of adjustment. The model says to monitor sectors, ordered by the computed fold hazard, and to monitor slopes rather than levels: the multiplier M_m , equivalently the local persistence of reallocation responses, rises toward infinity as the fold approaches, which is the standard critical-slowness diagnostic and here has a structural interpretation.

There is a second corollary worth stating because it is the security dilemma in equilibrium form. At a high-fragmentation equilibrium the hazard itself is higher: $\pi_m^{hi} - \pi_m^{lo} = \delta_m \theta (F_m^{hi} - F_m^{lo}) + \eta_a (F_m^{hi} - F_m^{lo}) > 0$. Firms fragment because they fear disruption, and their fragmentation makes disruption more likely as a fact about the adversary's escalation calculus once the hostage is spent. Both sides of the fold are rational-expectations equilibria. The fear is self-confirming in the high one.

3.3 Dynamics and hysteresis

Relationships are sticky: a fraction ρ of firms reconsider their sourcing each year, so the sector state evolves as

$$F_{m,t+1} = (1 - \rho) F_{m,t} + \rho T_m(F_{m,t}, F_t, \delta_t), \quad (7)$$

a monotone map in both the state and the hazard. Nothing locks a firm in: a reconsidering firm returns to cross-bloc sourcing whenever the threshold condition says to.

Proposition 4 (Temporary shocks, permanent networks). *Let the hazard follow a spike path: $\delta_t = \delta^{sp} > \delta_{hi}^*$ for T^{sp} periods, then $\delta_t = \delta^{term} \in (\delta_{lo}^*, \delta_{hi}^*)$ forever. There is a finite duration*

$\bar{T}(\delta^{sp}, \rho)$, decreasing in the overshoot and in the reconsideration rate, such that the economy converges to the high-fragmentation equilibrium of δ^{term} if $T^{sp} \geq \bar{T}$, and returns to the low one if $T^{sp} < \bar{T}$. Reversal from the high branch requires the hazard to fall below δ_{lo}^* , while the band $[\delta_{lo}^*, \delta_{hi}^*]$ is a region of history dependence.

The criterion inside the proof is simple enough to state: the spike must last long enough for the state to cross the *unstable* equilibrium of the terminal hazard, the separatrix. After that, the low equilibrium is unreachable under (7) even though every individual firm remains free to return. The trap is a property of the equilibrium set. Individual flexibility remains, and undoing the trap is a coordination problem rather than a compensation problem. Appendix C treats forward-looking firms, where inside the band expectations can select the branch, and gives the Krugman (1991) condition under which history wins.

The model is now complete as positive economics. Its normative content, who should pay whom at the margin, turns out to depend on where the sector sits relative to the fold, and that is the next section.

4 Welfare and Policy

4.1 The planner's margin and the sign of the wedge

Take a planner who can choose the sector's fragmented share, implementing it with threshold allocations, and who counts resource costs: premia actually paid, and disruption losses actually expected. Per unit of sector trade value,

$$B_m(F) = (1 - f_m^0)(1 - \chi F) \int_0^{q(F)} g dH_m + (1 - F) \pi_m(F) \lambda_m(F), \quad (8)$$

where $q(F)$ is the premium of the marginal switcher. Differentiating and evaluating at the decentralized equilibrium, where the marginal switcher is privately indifferent, the private terms cancel and what remains is the net external effect of one more exit:

$$\tau_m^*(F) = \underbrace{(1 - F)[\lambda_m(F) \delta_m \theta + \pi_m \lambda_m^0 \xi]}_{\text{hazard and loss imposed on stayers}} - \underbrace{\chi (1 - f_m^0) \int_0^{q(F)} g dH_m}_{\text{premium relief delivered to switchers}}. \quad (9)$$

Proposition 5 (The wedge and its sign flip). $\tau_m^*(F) > 0$ at low fragmentation whenever the deterrence channel operates ($\theta > 0$): the decentralized equilibrium has too much friend-shoring, and the corrective instrument is a subsidy of τ_m^* ad valorem on maintained cross-bloc trade. As $F \rightarrow 1$ with $\chi > 0$, $\tau_m^* < 0$: past sufficient fragmentation the capacity externality dominates and the planner subsidizes switching. A sign change therefore exists. Under the single-crossing condition verified at the calibration it is unique, at $\hat{F}_m$ near the fold.

The economics of (9) reads directly. A firm that exits ignores three things. It raises the escalation hazard on every remaining link: the deterrence externality, proportional to the trade

still standing. It thins the routes that spread disruption for those who stay. And it thickens the friendly supply base for those who have left. The first two argue that private friend-shoring is excessive, the third that it is insufficient, and which dominates depends on how much bridge is left to defend. Below the fold, most trade stands, the hostage is valuable, and the planner taxes exit. Beyond it, the hostage is spent, and the same planner accelerates what remains: defend the bridge, then burn it. This setting makes the corrective wedge on the same margin flip sign along a transition path, and Section 5 shows the flip is quantitatively real, from +6.4 percent at the post-2022 point to −4.9 percent at a dispute hazard of 24 percent.

4.2 Instruments

Three instruments in current use map cleanly into the model, and they operate on distinct margins.

A *friend-shoring subsidy* pays share ς of the switching premium. It shifts the threshold to $\hat{g}_m/(1 - \varsigma)$: mechanically it raises F_m , erodes the hostage, raises π_m , and drags both fold points toward the economy. It is the corrective instrument only where $\tau^* < 0$, past the flip. Deployed below the fold it has the wrong sign, and deployed near the fold it can tip the sector outright, a possibility Section 5 prices at the calibration.

A *stockpile or buffer* cuts the loss a disruption inflicts: λ_m^0 falls by the covered fraction. It lowers thresholds, lowers F_m , preserves deterrence, and pushes the folds away. In the model’s accounting it is the main resilience instrument because it purchases exactly the object firms fear, disruption cost, while preserving the object society needs, the hostage. Its resource cost prices the comparison in practice.

A *maintenance subsidy* of τ^* on cross-bloc trade in critical inputs implements the planner’s optimum directly. Its political economy is hard, subsidizing trade with the adversary in the name of security, and the model supplies the logic: the subsidy buys deterrence, which is a public good priced at zero by every private contract.

An independence motive enters cleanly. Let v be the government’s value per unit of trade rewired, the value of reduced dependence on a rival in an actual war. This motive subtracts from (9). The sign condition becomes $v < \tau^*$, so at the calibration the vulnerability motive must be worth more than 6.4 percent of trade value per year, every year, before zero maintenance subsidy becomes right. The same v raises the value of stockpiles symmetrically, since they deliver resilience while preserving deterrence.

5 Quantitative Analysis

5.1 Calibration

Table 1 collects the parameters and classifies each by source. The calculation is disciplined arithmetic, and the bifurcation diagrams report results as functions of the load-bearing objects.

Three entries carry the calculation. The losses λ_m^0 come from (1), so the critical sector’s loss of 1.50 (one and a half years of doubled input costs) is the arithmetic of $\sigma = 2$, a fifty percent cost

Table 1: Calibration (annual).

Parameter		Value	Category	Basis
Trade weights	w_m	0.15 / 0.35 / 0.50	calibrated sectors	critical / middle / flexible
Substitution elasticities	σ_m	2 / 4 / 8	estimated	HS-level span (BW)
Cross-bloc input shares	s_m	0.50 / 0.35 / 0.20	calibrated sectors	concentration of critical inputs
Disruption durations	D_m	1.5 / 1.0 / 0.5 yr	estimated	recovery times (case studies)
Implied losses (ad val. flow)	λ_m^0	1.50 / 15% / 1.6%	internal	equation (1)
Median premia	$\bar{g}_m$	0.20 / 0.15 / 0.10	calibrated input	Design 4
Premium dispersion	s	0.5	calibrated input	unit-value dispersion in Design 4
Pre-existing diversification	f_m^0	0.05 / 0.08 / 0.10	measured proxy	pre-2022 China-plus-one
Deterred share	θ	0.6 {0.4–0.7}	calibrated range	sign from MMT, Design 3
Thinning, capacity	ξ, χ	0.4, 0.4 {0–0.6}	calibrated range	Design 4
Systemic feedback	η_a	0.01	sensitivity input	small, spanned
Reconsideration rate	ρ	0.25	estimated	four-year links (BS)
Dispute hazard, pre-2022	δ^{pre}	4%	calibrated input	one dispute per 25 yr
Dispute hazard, post-2022	δ^{post}	10%	calibrated input	one per decade, full δ axis shown
Cross-bloc intermediates / GDP	α	4%	estimated	public trade aggregates

Notes: BW is Broda and Weinstein (2006), MMT is Martin et al. (2008), BS is Barrot and Sauvagnat (2016), and recovery case studies are Carvalho et al. (2021) and Boehm et al. (2019). Braces give the ranges used in the sensitivity exercises.

share, and an eighteen-month recovery, each drawn from the disruption literature. The premium distribution is load-bearing, and Design 4 estimates it from unit-value gaps. The medians here imply, through the model, a pre-2022 fragmented share of 5.0 percent in the critical sector, nearly all of it the non-geopolitical floor. The dispute hazards translate the 2022 reassessment into the model’s units: moving from one escalation-grade dispute per twenty-five years to one per decade places the economy on the δ axis, and every figure below traces the full axis.

5.2 Where the world sits: multipliers and the distance to the fold

The post-2022 steady state puts critical-sector fragmentation at 6.9 percent, against 5.0 before, with the equilibrium hazard on remaining critical links at 4.5 percent per year, of which 20 basis points is the feedback contribution: the erosion of deterrence by the switching that has already happened. The aggregate moves barely a point, from 8.6 to 8.8 percent: the reallocation is real, sectorally concentrated, and aggregate-invisible, which is the pattern the measurement literature reports (Gopinath et al., 2025; Freund et al., 2023).

Figure 1 shows the mechanism and Figure 2 the result. The critical sector’s fold sits at $\delta_{hi}^* = 13.1$ percent, with the hysteresis band beginning at 11.4: the move from 4 to 10 percent consumed about 66 percent of the pre-2022 distance to tipping. Across the deterrence and capacity grid of Table 5, the distance consumed runs from 47.0 to 100.0 percent. In the grid’s most fragile cell ($\theta = 0.4, \chi = 0.6$), the post-2022 hazard is already past the fold, so the critical sector’s question is timing. The multiplier at the current point is $M_c = 1.25$: cross-sectional estimates of post-2022 switching understate the equilibrium response by a quarter, and the gap diverges as δ approaches the fold. The middle and flexible sectors remain distant from their folds along the plotted axis. At $\delta = 25$ percent, a dispute every four years, the middle sector’s

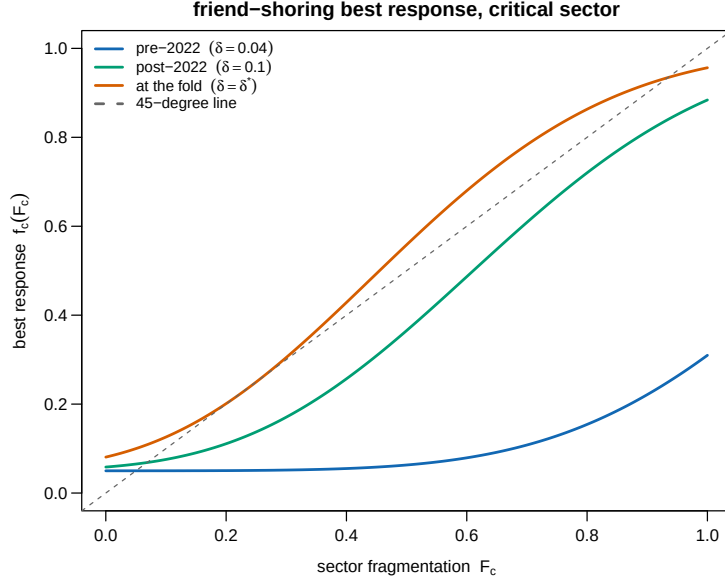


Figure 1: The critical sector’s best response at three dispute hazards. At the fold hazard δ^* the curve is tangent to the diagonal: the low equilibrium is destroyed.

fragmented share still sits at 8 percent, its non-geopolitical floor. Fragmentation, in this model, is a targeted amputation of specific limbs.

5.3 Hysteresis: a temporary confrontation, a permanent network

Figure 3 runs the paper’s central experiment. Two identical economies start at the post-2022 point. In one, the dispute hazard drifts up to 12.2 percent, inside the hysteresis band, and stays there: the critical sector settles at 12 percent fragmented. In the other, a confrontation raises the hazard to 20.1 percent, one escalation-grade dispute every five years, Cold War intensity, for twelve years, and then subsides to the same 12.2 percent: the sector converges to 90 percent fragmented and stays there forever. Same fundamentals at the end, a factor of seven between the outcomes, and every firm in both economies free, at every reconsideration, to source anywhere. Inside the trap the disruption hazard is 11.7 percent per year against 5.9 outside it: the confrontation’s permanent legacy is a rewired network and a world in which escalation is twice as likely, because the trade that deterred it is gone. The resource cost of the trap is 4.4 percent of the sector’s trade value per year, about 2.7 basis points of world GDP as accounted here. The accounting prices premia and expected disruption losses, with variety, diffusion, and political costs entering as additional measured margins in the extensions.

5.4 Policy at the calibration

Figure 4 prices the instruments of Section 4 at the calibration. The friend-shoring subsidy, paying 25 percent of the switching premium, moves the critical sector’s fold from 13.1 down to 9.7 percent: below the current dispute hazard. The sector tips: fragmentation goes to 94 percent,

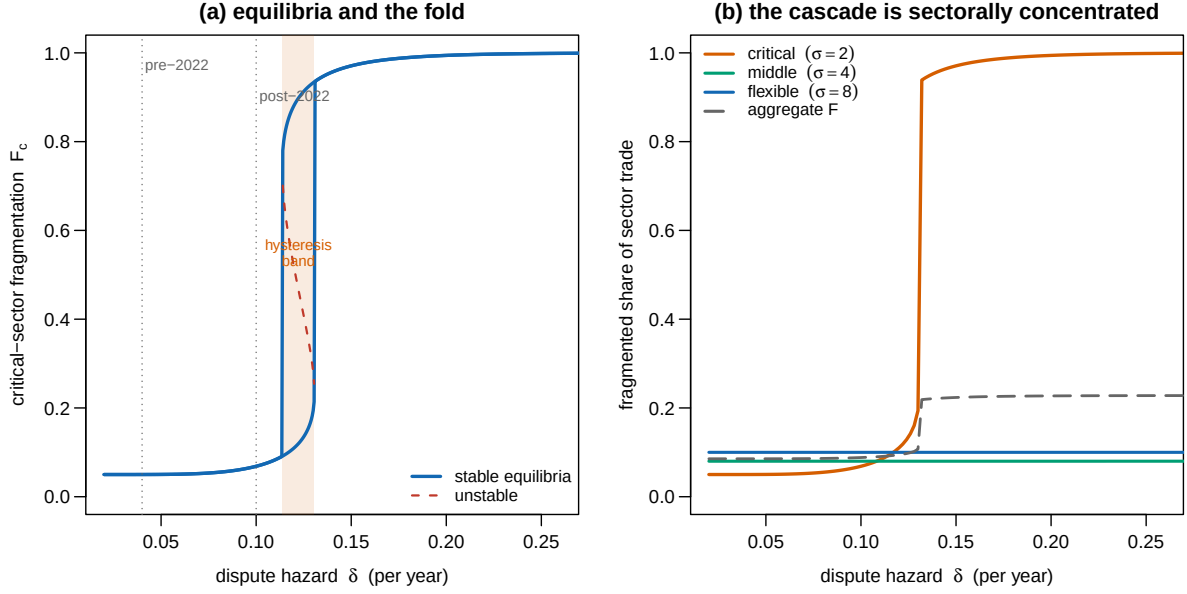


Figure 2: Panel (a): equilibria of the critical sector by dispute hazard. Solid branches are stable, dashed unstable, and the shaded region is the hysteresis band $[\delta_{lo}^*, \delta_{hi}^*] = [11.4\%, 13.1\%]$. Panel (b): fragmented shares by sector along the same axis, continuing from low initial conditions: the cascade is confined to the critical sector.

and the equilibrium disruption hazard rises from 4.5 to 9.9 percent per year. The subsidy, sold as insurance, more than doubles the probability of the event it insures against, because it spends the deterrent. The stockpile, cutting the disruption loss by a third, pushes the fold out to 19.7 percent, pulls fragmentation down to 5.2 percent, and lowers the hazard to 4.4: dominant on every margin the model prices. Figure 5 traces the corrective wedge itself: +6.4 percent ad valorem at the post-2022 point, rising through the band as the marginal hostage becomes more valuable, then flipping to -4.9 percent once the hazard is high enough that the planner, too, gives up the bridge. Sensitivity is where it should be: Appendix Table 5 shows the fold exists for every $\theta \geq 0.5$ once the capacity channel operates at $\chi \geq 0.2$, and disappears when both are shut, and Figure 6 draws the whole plane, which is precisely why Design 4's estimate of χ is the single most consequential number the empirical program can produce.

6 Empirical Designs

The model's quantitative claims rest on five objects: the direct switching elasticity, the deterrence share θ , the thinning and capacity elasticities ξ and χ , and the premium distribution H_m . Each gets its own design and its own data, and Corollary 1 dictates the architecture: cross-sectional designs deliver direct effects, and the feedback parameters come from moments that vary the aggregate. The designs below state the predicted signs, magnitudes, and estimating samples.

Design 1: the direct effect, by criticality. Product-level bilateral trade from BACI (CEPII,

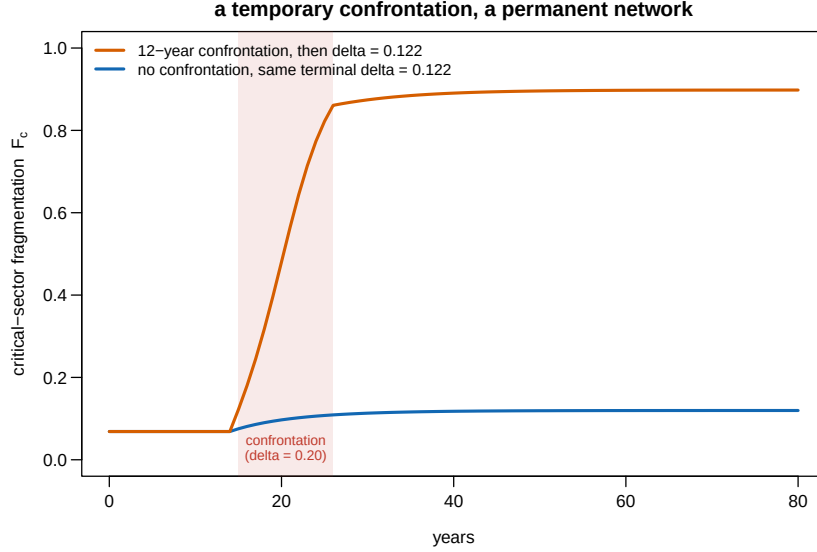


Figure 3: Two economies facing the identical terminal dispute hazard $\delta = 12.2\%$, inside the band. One lived through a twelve-year confrontation at $\delta = 20.1\%$. The other entered directly at the terminal hazard.

2026) at HS-6, 2015–2024, in a triple-difference event study around February 2022:

$$\Delta \ln \text{Share}_{ijm,t} = \alpha_{im} + \alpha_{jm} + \tau_{mt} + \sum_k \beta_k \mathbf{1}\{t = t_0 + k\} \times \text{GeoDist}_{ij} \times \lambda_m^0 + \Gamma' X_{ijmt} + \varepsilon_{ijmt}, \quad (10)$$

where GeoDist_{ij} is the pre-2022 ideal-point distance from United Nations votes (Bailey et al., 2017) and λ_m^0 is constructed, before the event, from published substitution elasticities (Broda and Weinstein, 2006), input-output cost shares (OECD, 2026), and disruption-duration proxies. The model makes three predictions. The coefficients β_k are negative, grow along the adjustment path, and persist. The critical-versus-flexible differential is about 1.9 percentage points of the cross-bloc share at the post-2022 steady state. And, most sharply, the response is *dose-monotone* in λ_m^0 : the interaction with a constructed criticality index, rather than a sector dummy, separates the mechanism from sector-specific policy shocks. Controls absorb tariffs, freight, and commodity cycles, and the 2018–19 tariff episode supplies a placebo period in which trade moved while dispute hazards stayed fixed.

Design 2: the extensive margin and the hysteresis signature. Exit and re-entry of supply relationships at the product-country level, in hazard form. Two predictions distinguish the model from a smooth repricing story. First, exits concentrate where (5) says: high λ_m^0 , high geopolitical distance, low premium. Second, and diagnostic: *exited relationships return slowly when tensions ease*. A transitory-shock model implies re-entry symmetric to exit. Proposition 4 implies an asymmetry that grows with the sector’s proximity to its fold. The comparison group is exits induced by the 2018 tariffs, which the tariff literature finds partially reversed and rerouted (Freund et al., 2023; Alfaro and Chor, 2023): geopolitical exits should be stickier than tariff exits, conditional on the same cost fundamentals.

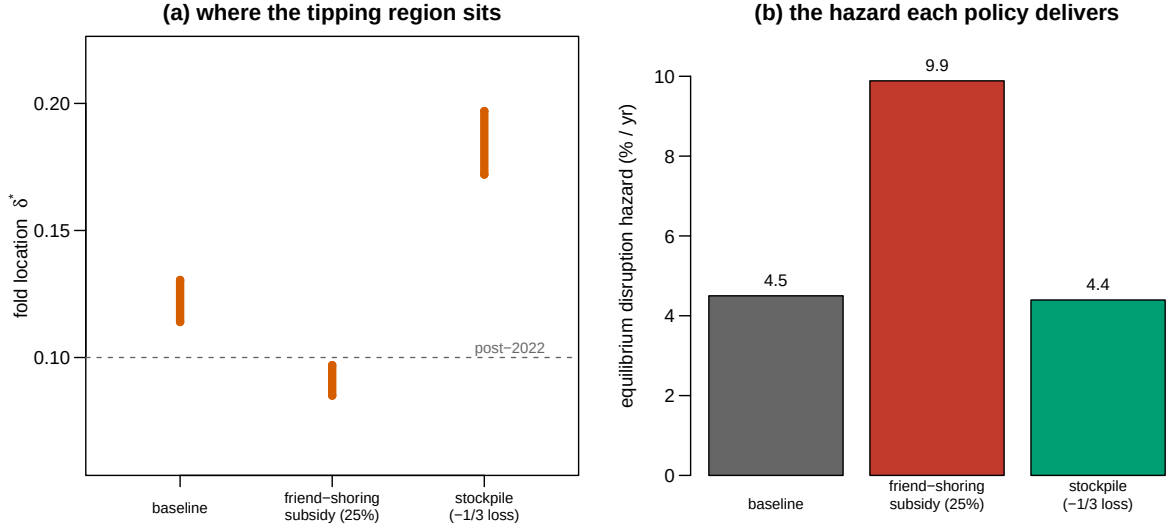


Figure 4: Panel (a): the tipping region under three policies. The dashed line is the post-2022 dispute hazard. Panel (b): the equilibrium disruption hazard each policy delivers at $\delta = 10\%$.

Design 3: the deterrence share θ . The model’s engine is measurable on its own: equation (2) says the probability that a dispute escalates into trade disruption falls with the trade still crossing the boundary in the targeted sector. The Global Sanctions Data Base (Felbermayr et al., 2020), restricted to export controls and supply-touching measures and combined with a dispute chronology, gives the left-hand side, and remaining cross-bloc coverage $(1 - F_m)$, measured from BACI, the right:

$$\Pr(\text{escalation}_{mt}) = \Lambda(\alpha_m + \tau_t - \theta_E (1 - F_{m,t-1}) + \Xi' Z_{mt}), \quad (11)$$

the Martin et al. (2008) relation at the sector level. The identifying variation comes from pre-period criticality λ_m^0 interacted with aggregate reassessment shocks. That instrument shifts coverage through firms’ sourcing margins while leaving the government’s contemporaneous benefit b to be absorbed by dispute, sector, and time controls.

Design 4: the feedback parameters and the distance to the fold. The premium distribution comes from unit-value gaps between in-bloc and cross-bloc suppliers of the same HS-6 product, pre-2022: location $\bar{g}_m$ and dispersion s read off directly. The capacity elasticity χ comes from the compression of those gaps as cumulative reallocation proceeds: the post-2018 Vietnam and Mexico expansions are the identifying episodes. The thinning elasticity ξ comes from disruption-duration data conditional on corridor volume. With $(\bar{g}_m, s, \chi, \xi, \theta)$ in hand, the slope Φ'_m and the fold hazards are calculations, and the paper’s bottom line becomes an estimate with a standard error: the distance to the critical sector’s tipping point, and the multiplier M_m that converts every reduced-form estimate in Designs 1 and 2 into its equilibrium counterpart. Since the fold’s existence turns on χ (Table 5), the unit-value compression estimate is the single

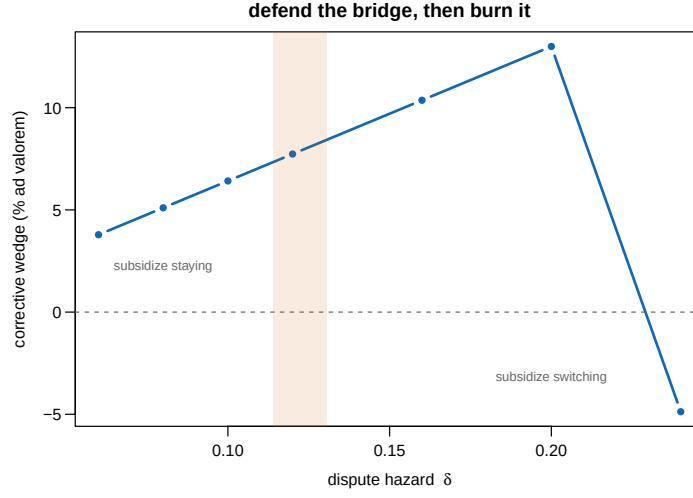


Figure 5: The corrective wedge τ^* by dispute hazard: positive (subsidize maintained cross-bloc trade) below and inside the band, negative (subsidize switching) beyond it.

most consequential moment in the program.

Table 2 collects the tests, and two carry the theory. P4 measures the engine: the response of escalation probabilities to remaining trade pins down the deterrence externality and the first term of the wedge (9). P3 measures the equilibrium structure: hysteresis in relationships, beyond what sunk costs explain, is the fingerprint that separates fold dynamics from smooth repricing. Together the two tests locate the complementarity between security deterrence and market capacity.

Two facts line up with the model before entering the calibration. The calibration uses elasticities, cost shares, durations, and the hazard translation. The sectoral pattern of the post-2022 reallocation, concentrated in low-elasticity products, is documented by Gopinath et al. (2025) and is exactly where the model places the response. The aggregate smallness of the move is also a prediction of Proposition 3(iii), since a sectoral fold can pass through aggregate statistics quietly.

Every dataset in the empirical program is public. Table 3 records the implementation.

7 Conclusion

I have asked why trade fragmentation might feed on itself, and answered with a mechanism old enough to trust: trade deters the disruption of trade, so the anticipatory rewiring of supply chains consumes the very deterrent that made cross-bloc sourcing safe. Closing that loop turns the fragmentation literature’s exogenous exercises into equilibrium objects with sharp properties: a multiplier that cross-sectional evidence systematically misses, tipping points ordered by a computable criticality statistic, hysteresis with free re-entry, and a corrective wedge that starts positive, favoring the maintenance of cross-bloc trade, and flips sign only when the bridge has lost its defense value. At the calibration, the 2022 reassessment moved the critical sector two

Table 2: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
P1: cross-bloc shares fall after 2022, by geopolitical distance	−1.9 pp critical vs. flexible	0	BACI triple difference
P2: response dose-monotone in constructed criticality λ_m^0	ordering of (1)	flat	criticality interactions
P3: geopolitical exits stickier than tariff exits	asymmetry rising near fold	symmetry	relationship hazards
P4: escalation hazard falls with remaining sector coverage	$\theta \approx 0.6$	$\theta = 0$	sanctions-dispute panel
P5: premium gaps compress with cumulative reallocation	$\chi \approx 0.4$	$\chi = 0$	unit-value gaps
P6: reallocation persistence rises as folds approach	$M_m \uparrow$, critical slowing down	constant	sector dynamics

Notes: magnitudes at the baseline calibration. Data: CEPII BACI, OECD ICIO, UN ideal points, Global Sanctions Data Base, published HS-level substitution elasticities.

thirds of the way to its fold, and between half and all of the way across the sensitivity grid. A friend-shoring subsidy of the scale now fashionable covers the rest of the distance and doubles the disruption hazard it was sold to insure against. A stockpile of equal ambition does the opposite. Section 6 specifies the moments that turn these calibrated statements into estimates, with Design 4 delivering the standard error on the distance to the tipping point.

The result I would defend most broadly is the sign of the wedge. The current policy consensus treats private supply chains as insufficiently alarmed and subsidizes their retreat. In this model the private sector is, if anything, excessively alarmed at the margin, because each firm omits the deterrence its trade provides to everyone else: security, here, is a public good produced jointly with commerce, and the market underprovides commerce for precisely the reason markets underprovide public goods. Whether the deterrence share is 0.6 or 0.2 is an empirical question with a specified design. A positive deterrence effect is one of the better-established regularities in the political economy of conflict. Policy built on the vulnerability intuition alone prices that effect at zero.

Three extensions seem productive. Connector countries, which reroute trade rather than end it, scramble the mapping between measured bilateral fragmentation and the hostage that matters, and Appendix F shows their net effect on tipping is genuinely ambiguous. Endogenizing the dispute hazard’s arrival as well as its escalation would close a second loop, from fragmentation to the temperature of the rivalry itself. And the measurement is the point: the model was built so that each of its dangerous parameters is a coefficient in a feasible regression, and the data to run them are public.

Table 3: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or proxied	Measure
CEPII BACI	HS-6 bilateral, 1995–2024	annual	public	observed	in hand
OECD ICIO	cost shares, centrality	annual	public	observed	in hand
HS-level elasticities	Broda and Weinstein (2006)	fixed	published	estimated, period	pre- in hand
Disruption durations	case literature	episode	published	proxied	to assemble
GSDB export controls	1950–2022	episode	public	observed	in hand
UN ideal points	all members	annual	public	observed	in hand
Unit values (premia)	BACI val-ues/quantities	annual	public	proxied, noise	known in hand

Notes: unit values are the known weak link, noisy at HS-6 and sensitive to composition. Design 4 therefore uses within-product changes and the post-2018 reallocation episodes rather than levels.

A Proofs

A.1 Proof of Lemma 1

Let $V^{sw} = -g^{\text{eff}}/(1 - \beta_f)$ be the value of the absorbing friend-shored state and V^{st} the value of staying, which pays the expected loss and transits to the absorbing state upon disruption: $V^{st} = -\pi\lambda + \beta_f[(1 - \pi)V^{st} + \pi V^{sw}]$. Solving, $V^{st}(1 - \beta_f(1 - \pi)) = -\pi\lambda + \beta_f\pi V^{sw}$, and $V^{st} \geq V^{sw}$ iff $-\pi\lambda + \beta_f\pi V^{sw} \geq (1 - \beta_f(1 - \pi))V^{sw}$ iff $-\pi\lambda \geq (1 - \beta_f)V^{sw} = -g^{\text{eff}}$, which is the flow comparison. No discount-rate term survives. ■

A.2 Deterrence microfoundation

Lemma 2 (Linear hazard from escalation choice). *Suppose disputes arrive with probability δ_m , and conditional on a dispute the adversary escalates iff a benefit $b \sim U[0, \bar{b}]$ exceeds the destroyed surplus $\vartheta_m \bar{S}_m(1 - F_m) + \vartheta_a \bar{S}(1 - F)$. Then the link hazard is affine and increasing in (F_m, F) , and equals (2) up to a reparametrization of (δ_m, θ) that normalizes the hazard at zero fragmentation.*

Escalation probability is $1 - [\vartheta_m \bar{S}_m(1 - F_m) + \vartheta_a \bar{S}(1 - F)]/\bar{b}$ on the interior. Multiplying by δ_m and collecting the constants into the normalized pair gives (2). Any log-concave benefit distribution preserves monotonicity in both arguments. Uniformity buys exact linearity only. ■

A.3 Proof of Proposition 1

Write

$$\hat{g}_m(F_m, F) = \delta_m \lambda_m^0 \frac{(1 - \theta) + \theta F_m + \eta_a F}{1 - \chi F_m} (1 + \xi F_m).$$

Holding F fixed, the product and quotient rules give

$$\frac{\partial \hat{g}_m}{\partial F_m} = \delta_m \lambda_m^0 \left[\frac{\theta(1 + \xi F_m)}{1 - \chi F_m} + \frac{\xi[(1 - \theta) + \theta F_m + \eta_a F]}{1 - \chi F_m} + \frac{\chi[(1 - \theta) + \theta F_m + \eta_a F](1 + \xi F_m)}{(1 - \chi F_m)^2} \right].$$

The three terms are the deterrence, thinning, and capacity channels in the proposition, and each is strictly positive when its parameter is positive. Holding F_m fixed gives

$$\frac{\partial \hat{g}_m}{\partial F} = \frac{\eta_a \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m} > 0$$

when $\eta_a > 0$. With $\theta = \xi = \chi = \eta_a = 0$, $\hat{g}_m = \delta_m \lambda_m^0$ is constant, T_m is independent of (F_m, F) , and (6) has the unique solution $F_m = T_m$, linear in threshold shifters through H_m . ■

A.4 Proof of Proposition 2

T_m is continuous and increasing in F_m , with $T_m(f_m^0) \geq f_m^0$ and $T_m(1) \leq 1$, so $R(F) = T_m(F) - F$ has a zero by the intermediate value theorem. For smooth H_m , transversal roots satisfy $R'(F) = T'_m(F) - 1 \neq 0$. Transversality is the generic case after any small change in a primitive parameter,

and it gives a finite number of roots on compact subintervals. Since R is nonnegative at the lower boundary and nonpositive at the upper boundary, the sign of R alternates across transversal roots. Stable roots have $T'_m < 1$, unstable roots have $T'_m > 1$, and the first and last roots are stable when they are interior. Implicit differentiation of $F^* = T_m(F^*, z)$ gives

$$\frac{dF^*}{dz} = \frac{\partial T_m / \partial z}{1 - \Phi'_m}, \quad \Phi'_m = T'_m(F^*), \quad \frac{\partial T_m}{\partial z} = (1 - f_m^0)h_m(\hat{g}_m) \frac{\partial \hat{g}_m}{\partial z}.$$

As a fold is approached along the stable branch, $\Phi'_m \uparrow 1$ and the equilibrium response diverges. ■

A.5 Proof of Corollary 1

Within an equilibrium, (F_m, F) are fixed. Variation across units i in any idiosyncratic shifter (exposure, distance, premium) moves individual thresholds and leaves the common feedback term fixed, so a regression of choices on shifters, however identified, estimates $(1 - f_m^0)h_m \partial \hat{g} / \partial z$: the numerator of Proposition 2's ratio. The denominator $1 - \Phi'_m$ requires variation in (F_m, F) themselves, which within-equilibrium comparisons hold fixed. ■

A.6 Proof of Proposition 3

(i) T_m is increasing in δ pointwise, so extremal fixed points are increasing in δ by monotone comparative statics. In the S-shaped case described by Proposition 2, with transversal roots, the condition $\max_F \Phi'_m > 1$ gives an interval on which T_m crosses the diagonal from below. The boundary signs then force three crossings: a low stable root, an unstable root, and a high stable root. As δ rises, the graph of T_m moves up pointwise. At δ_{hi}^* the low stable root and the unstable root coalesce at $\Phi'_m = 1$ and disappear, so the extremal-low selection jumps to the high branch. The lower fold δ_{lo}^* is the symmetric coalescence of the high and unstable roots as δ falls. (ii) With $\eta_a = 0$ and the other non-loss primitives fixed, (5) depends on (δ, λ_m^0) only through the product $\delta \lambda_m^0$. Hence the entire fixed-point correspondence, including the folds, is a function of $\delta \lambda_m^0$. Fold hazards satisfy $\delta_m^* \lambda_m^0 = \text{constant}$ across sectors sharing (H, θ, ξ, χ) , exactly inverse to criticality. (iii) Immediate from $F = \sum_m w_m F_m$. ■

A.7 Proof of Proposition 4

The map $F \mapsto (1 - \rho)F + \rho T(F, \delta)$ is increasing in F and δ , so paths from ordered initial conditions stay ordered, and with constant δ the path is monotone toward the nearest fixed point in its direction of motion. During the spike, $\delta^{sp} > \delta_{hi}^*$ implies a unique fixed point $F^{hi}(\delta^{sp})$ exceeding the terminal unstable point $U(\delta^{term})$. The path rises toward it and crosses $U(\delta^{term})$ at some finite $\bar{T}$, decreasing in δ^{sp} (which raises the map) and in ρ (which speeds it). If reversion occurs after $\bar{T}$, the state exceeds $U(\delta^{term})$, where $T(F, \delta^{term}) > F$ holds on (U, F^{hi}) : the path rises monotonically to the high branch. If before, the state is below U , where $T < F$: it falls back. From the high branch, δ must fall below δ_{lo}^* , where the high fixed point is destroyed, before any monotone path can leave its basin. ■

A.8 Proof of Proposition 5

Differentiate (8) in F , using $(1 - f_m^0)h_m(q)q'(F) = 1$. The premium term gives

$$\frac{d}{dF} \left[(1 - f_m^0)(1 - \chi F) \int_0^{q(F)} g dH_m \right] = (1 - \chi F)q(F) - \chi(1 - f_m^0) \int_0^{q(F)} g dH_m.$$

The exposure term gives

$$\frac{d}{dF} [(1 - F)\pi_m(F)\lambda_m(F)] = -\pi\lambda + (1 - F) [\lambda \delta_m \theta + \pi \lambda_m^0 \xi].$$

At the decentralized equilibrium the marginal switcher is indifferent, $(1 - \chi F)q(F) = \pi\lambda$, so the private terms cancel. Therefore

$$\frac{dB_m}{dF} = (1 - F) [\lambda \delta_m \theta + \pi \lambda_m^0 \xi] - \chi(1 - f_m^0) \int_0^{q(F)} g dH_m = \tau_m^*(F).$$

A positive τ_m^* means one more exit raises social burden, so decentralized exit is excessive and the corrective instrument subsidizes maintained trade. At $F = f_m^0$ the integral vanishes and $\tau^* = (1 - f_m^0)[\lambda \delta \theta + \pi \lambda^0 \xi] > 0$ when $\theta > 0$. As $F \rightarrow 1$ the first bracket vanishes while the integral tends to $(1 - f_m^0)\mathbb{E}[g | g < q(1)] > 0$, so $\tau^* < 0$ when $\chi > 0$. Continuity gives a crossing. Uniqueness holds under the single-crossing condition verified at the calibration. A per-period ad valorem payment τ to maintained trade shifts the private comparison to $\pi\lambda - \tau \leq g^{\text{eff}}$, and therefore spans all threshold allocations. ■

B Microfoundations

The CES loss. Let the firm's input bundle be CES over the cross-bloc variety and a composite substitute, elasticity σ_m , with the variety's cost share s_m at initial prices. Removing the variety multiplies the bundle's price index by $(1 - s_m)^{1/(1-\sigma_m)}$: with $\sigma_m = 2$ and $s_m = 1/2$, a doubling, and with $\sigma_m = 8$ and $s_m = 0.2$, a three percent rise. A disruption lasting D_m years, before replacement contracts are in place, costs the flow equivalent (1). Downstream propagation through input-output rounds scales λ_m^0 by a centrality factor (Baqaee and Farhi, 2019; Carvalho et al., 2021). The baseline reports the direct sectoral loss, and the centrality extension prices the multiplier separately.

The escalation game. Lemma 2 treats one dispute. The same argument applies to repeated disputes so long as the escalation choice is static ex post, which it is here because portfolios are predetermined within the period. A forward-looking adversary, for whom today's escalation accelerates tomorrow's fragmentation and erodes future hostage-taking capacity, adds a second time-inconsistency problem on the government side. The present specification keeps the demand side clean and puts all strategic complementarity in firms' sourcing choices.

Capacity scale. The compression factor $(1 - \chi F_m)$ stands for entry and learning in the friendly supply base: with an in-bloc supply curve of elasticity $1/\chi$ facing cumulative demand proportional to F_m , the premium of the marginal in-bloc supplier falls linearly in F_m to first order. The identifying content is that the compression loads on *cumulative* reallocation rather than contemporaneous volumes, which is what Design 4 exploits.

C History versus Expectations

The dynamics (7) are adaptive: reconsidering firms take the current state as their forecast. With forward-looking firms, a reconsidering firm compares present values along an anticipated equilibrium path, and inside the band $[\delta_{lo}^*, \delta_{hi}^*]$ there can exist self-fulfilling paths to either branch: the overlap of Krugman (1991) and Matsuyama (1991). The scope for expectations to override history is governed by the race between discounting and adjustment: when the reconsideration rate is slow relative to the discount rate, the future path matters little to the marginal chooser and history determines the outcome, and when firms are patient and adjustment fast, an anticipated migration can pay for itself. Lemma 1 sharpens the comparison here. Because continuation values cancel in the stationary comparison, the option term along a transition is proportional to the anticipated *change* in thresholds over the firm’s expected commitment spell, of length $1/\rho$, so the adaptive benchmark is exact in stationary environments and errs along transitions by that term. A full perfect-foresight treatment, including whether announcements alone can spring the trap of Proposition 4, is the natural next paper. The policy analysis of Section 4 is conservative in the meantime because coordination instruments only add to the planner’s kit.

D Parameter Discipline and Sensitivity

Table 4 records, for every parameter group, where it enters the model, its units, its source, the range examined, and what moves when it moves. The exogenous-risk benchmark obtains at $\theta = \xi = \chi = \eta_a = 0$ (Proposition 1). In the sensitivity grid, the fold appears for $\theta \geq 0.5$ with $\chi \geq 0.2$ or for $\theta = 0.7$ alone (Table 5), with premium dispersion below roughly 0.7 log points (Section 3). The wedge’s sign flip is governed by $\chi > 0$.

E Data Appendix

Trade. CEPII BACI, HS-6 bilateral flows and unit values, 1995–2024 (CEPII, 2026), with extensive-margin relationship panels constructed at exporter-importer-product level. Bloc and alignment measures from UN ideal points (Bailey et al., 2017), fixed at pre-2022 values.

Criticality. Substitution elasticities at HS level from Broda and Weinstein (2006), cost shares and downstream centrality from OECD ICIO (OECD, 2026), and disruption durations proxied from the recovery literature (Carvalho et al., 2021; Boehm et al., 2019). The index λ_m^0 is computed entirely from pre-period data.

Table 4: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source	Range examined	Effect of the range
σ_m, s_m, D_m	eq. (1)	elasticity, share, years	Broda and Weinstein (2006), ICIO, recovery studies	spanned across sectors	fix λ_m^0 , hence tipping order
λ_m^0	eq. (5)	ad val. flow	implied by eq. (1)	1.6% to 150%	fold position scales as $1/\lambda_m^0$
$\bar{g}_m, s$	eq. (6)	ad val. orem, log pts	calibrated input, Design 4	s 0.3 to 1.1	fold gone above $s \approx 0.7$
f_m^0	eq. (6)	share	pre-2022 diversification	floor check	level floor only
θ	eq. (2)	share de-terred	sign from Martin et al. (2008), Design 3	0.4 to 0.7	fold hazard 9.9 to 17.1%, consumed distance 47.0 to 100.0%
ξ, χ	eqs. (3), (4)	elasticities	calibrated input, Design 4	χ 0 to 0.6	fold existence (Table 5), wedge flip needs $\chi > 0$
η_a	eq. (2)	per year	sensitivity input	0 to 0.03	minor at calibration
ρ	eq. (7)	per year	Barrot and Sauvagnat (2016)	timing check	timing of $\bar{T}$ only
$\delta^{pre}, \delta^{post}$	eq. (2)	per year	calibrated input	full axis in figures	positions the economy on every diagram
w_m, α	aggregation	shares	trade aggregates	scale check	scale GDP numbers

Notes: the criticality inputs (σ_m, s_m, D_m) are estimated objects with published counterparts, and Designs 3 and 4 estimate the feedback elasticities (θ, ξ, χ). Every headline number reports its sensitivity to the feedback block.

Disputes and escalations. The Global Sanctions Data Base (Felbermayr et al., 2020) restricted to export controls and supply-touching financial measures, augmented with a chronology of export-control announcements (the October 2022 United States semiconductor controls being the archetype). Escalation events are coded at the sector-month level.

Fragmentation. $F_{m,t}$ is measured as the share of sector m ’s pre-2022 cross-bloc trade value in sustained relationship exits, from the BACI extensive margin: the empirical counterpart of the model’s state variable.

F Extensions

Connector countries. Let a share of “friend-shoring” reroute through politically intermediate countries that re-export the adversary’s content. Connectors lower the effective premium (they are the cheap safe supplier), which raises fragmentation of direct links. But if rerouted trade still generates surplus that escalation would destroy, the hostage erodes less than measured fragmentation suggests. The net effect on the fold is ambiguous and empirically resolvable: it turns on whether escalation calculi price direct trade or total content, testable in Design 3 by instrumenting coverage under direct-trade and total-content exposure measures (Gopinath et al.,

2025; Freund et al., 2023).

Two-sided fragmentation. Making both blocs' firms choose adds the adversary's exits to the hostage erosion. With symmetric parameters the sector map's slope doubles where deterrence is concerned, moving folds closer. The one-sided model is thus conservative about tipping.

Correlated disruption and percolation. Here a severed link harms only its firm. In production networks link shocks propagate (Elliott et al., 2022; Baqaee and Farhi, 2019), making λ depend on the pattern as well as the level of fragmentation. The interaction of strategic hazard with percolation fragility is unexplored and, I suspect, reinforcing.

The vulnerability motive. If the government values rewired trade at v per unit beyond firms' losses, the wedge becomes $\tau^* - v$ and the planner's problem otherwise stands. All sign results survive for $v < \tau^*$, and the policy discussion in Section 4 prices the threshold.

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Table 5: Sensitivity: the critical sector’s fold band $[\delta_{lo}^*, \delta_{hi}^*]$ by deterrence share θ and capacity elasticity χ , in dispute-hazard units. “Unique” means a unique equilibrium at all hazards.

	$\chi = 0.0$	$\chi = 0.2$	$\chi = 0.4$	$\chi = 0.6$
$\theta = 0.4$	none	none	0.107–0.107	0.086–0.099
$\theta = 0.5$	none	none	0.113–0.117	0.087–0.110
$\theta = 0.6$	none	0.138–0.141	0.116–0.133	0.088–0.127
$\theta = 0.7$	0.165–0.171	0.144–0.163	0.119–0.157	0.090–0.151

Figure 6 refines Table 5 on a grid of 165 calibrations. The fold region is a northeast wedge: the two feedbacks substitute along a downward-sloping existence boundary, so the fold appears at $\chi \geq 0.15$ when $\theta = 0.6$ and requires zero capacity channel once θ reaches 0.65. Of the 165 cells, 48 are monotone everywhere, 117 carry a fold, and 16 have it below the post-2022 hazard already, all of them the neighborhood of the fragile corner of the coarser table. The dangerous region is the northwest of the wedge, weak deterrence with a strong capacity feedback, where the fold sits at low hazards. At $\theta \leq 0.35$ the plane passes from monotone directly to past-fold as χ rises, and the intermediate warning band disappears: for those calibrations the first sign of the fold would be the jump itself.

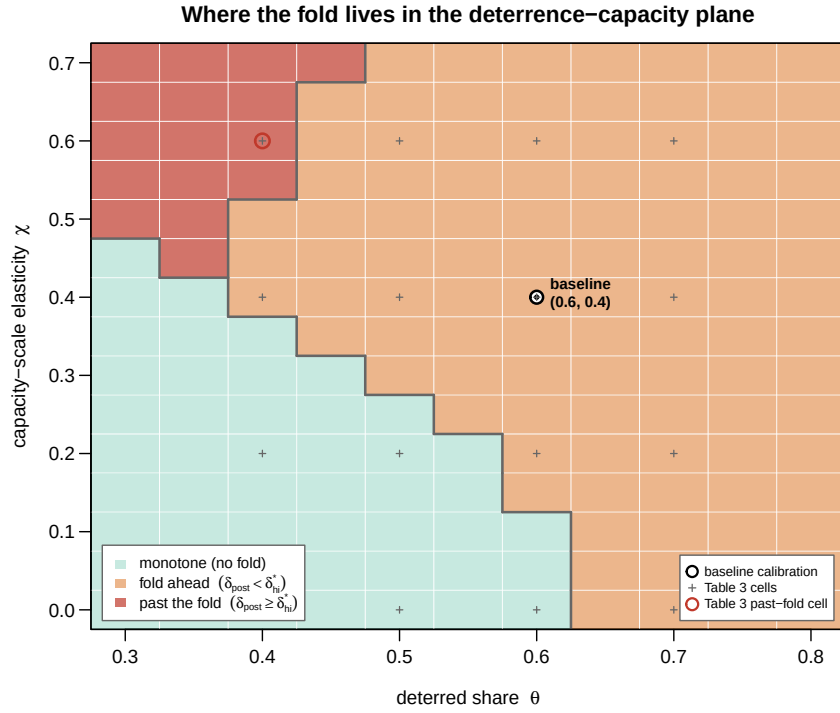


Figure 6: The fold in the (θ, χ) plane at the calibrated dispersion. Shading classifies each cell: unique equilibrium at every hazard, fold ahead of the post-2022 hazard, or fold already below it. Crosses mark the cells of Table 5, the dot marks the baseline, and the circled cell is the fragile calibration discussed in the text.